

By thevenin theorem;

$$I_L = \frac{V_{th}}{(R_{th} + R_L)}$$

Apply KVL, $10 = 470 I + 470 I$.

$$I = 0.01 A$$

using voltage division rule;

$$V_{th} = V_L \times \frac{R_1}{R_1 + R_2}$$

$$V_{th} = 10 \times \frac{470}{470 + 470} = 5V$$

to find thevenin resistance:-

w.k. $R_1 \parallel R_2$

$$R_p = \frac{470 \cdot 470}{2(470)} = 235 \Omega$$

$$R_{th} = R_{560} + R_p = 795 \Omega$$

According to thevenin theorem;

$$I_L = \frac{V_{th}}{R_{th} + R} = \frac{5}{1000 + 795} = 0.00275$$

$$I_L = 2.75 mA$$